A distributed maturation delay in the profit–investment cycle: differentiable inverse calibration and identifiability limits

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June 7, 2026

Abstract

We study the profit–investment cycle of the U.S. economy (quarterly, 1948–2026) through the lens of overaccumulation: past investment depresses current profitability with a delay. Using the cross-correlation function and a recession event study, we show a dominant contemporaneous co-movement together with a negative feedback centred about one year, which builds up before recessions—consistent with Tapia Granados (2012). Fixed-phase predator–prey oscillators (Lotka–Volterra, FitzHugh–Nagumo) cannot reproduce this shape. We instead pose a physical *maturation-chain* model, a distributed-delay ODE, and calibrate it by a differentiable inverse (physics-informed neural network with Fourier features). The inverse recovers the maturation lag cleanly on synthetic data (8.9% error) and estimates ≈ 1 year on real data; the lag is stable to transform choice and leave-one-crisis-out. Our central result is one of identifiability: the *centre* of the delay is robustly identified, but its *distributional width* is not, and the feedback is observationally equivalent to a linear VAR($p \geq 4$) at the level of the second-order autocovariance.

1 Introduction

The Marxian tradition of overaccumulation (Astarita; Tapia Granados, 2023) holds that capital accumulation eventually depresses the rate of profit, generating endogenous cycles and crises. A long line of nonlinear models—starting from Goodwin's (1967) growth cycle, a Lotka–Volterra (LV) predator–prey system, and its empirical predator–prey estimate by Goldstein (1999)—formalises this as an antagonism between profits and a second variable. These models impose an *instantaneous* coupling with a fixed 90° phase lag.

We ask a narrower, mechanistic question: is the feedback from investment to profits an instantaneous antagonism, or a *delayed* one whose timing reflects the maturation (gestation) of investment? We find the latter. The empirical signature is a contemporaneous co-movement plus a negative feedback concentrated about one year out—a shape no fixed-phase oscillator can generate, but a distributed-delay system can. We formalise this as a physical maturation-chain ODE and calibrate it by a differentiable inverse problem. The contribution is descriptive and methodological: a transform-robust, leave-one-out-stable estimate of a structural maturation lag, recovered by an inverse method where classical single-shooting fails, together with an explicit characterisation of *what cannot* be identified from these data.

2 Data and methods

We use quarterly U.S. corporate profits and gross private domestic investment (FRED, 1948–2026, $n = 313$). To avoid the spurious order-4 structure that the year-over-year transform injects at the lag of interest, we work in quarter-over-quarter growth, $x_t = \Delta \log$, which we verified to roughly halve the apparent feedback magnitude relative to YoY.

Descriptive. We compute the sample cross-correlation function (CCF) $\rho(k) = \mathrm{corr}(\mathrm{profits}_t, \mathrm{investment}_{t+k})$ and, around the twelve NBER recessions, a superposed-epoch (event-study) composite aligned at the investment trough.

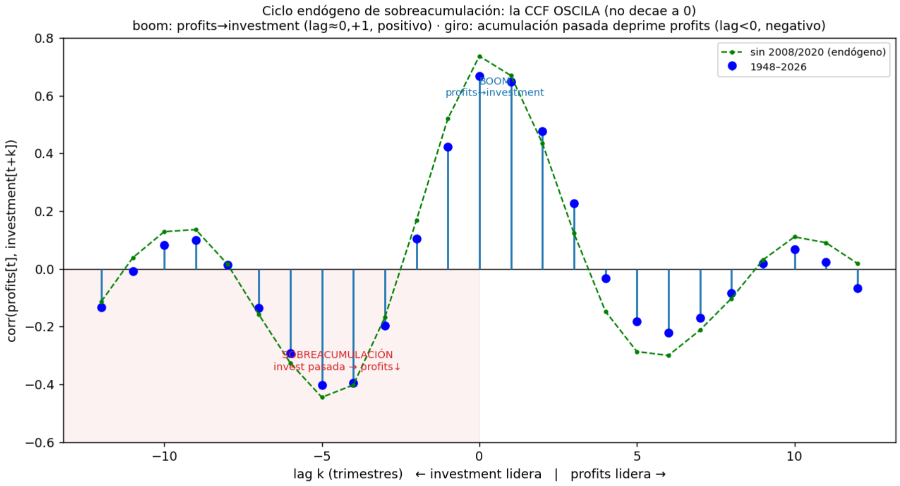

Figure 1: Cross-correlation profits↔investment. The function oscillates: a strong contemporaneous co-movement and a delayed negative feedback (investment maturing ∼1 year earlier depresses current profits). Robust to excluding 2008/2020.

Physical model. Investment matures through n stages before depressing profits; by the linear-chain trick this realises a Gamma-distributed delay of mean μ:

$$\dot{P} = c\,I - d\,P - k\,m, \qquad \dot{I} = a\,P - b\,I, \qquad m(t) = \textstyle\sum_j w_j(\mu)\,I(t-j), \tag{1}$$

with P profits, I investment, m matured investment, and $w(\mu)$ a Gamma kernel of mean μ (the *maturation lag*).

Differentiable inverse. We calibrate (a, b, c, d, k, μ) by an inverse physics-informed neural network (PINN): a network $t \mapsto [P, I]$ trained with a data loss on the observations and a physics-residual loss enforcing the ODE, optimising network weights and ODE parameters jointly (loss $= \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{phys}}$, $\lambda = 1$; after Raissi et al., 2019; Rackauckas et al., 2020). On long windows a plain network suffers spectral bias and the lag collapses; *Fourier-feature* inputs remove this. We validate recovery on synthetic data with a known lag, and assess identifiability by a profile over μ, a leave-one-crisis-out analysis, and the theoretical CCF of a fitted VAR(p) obtained from the discrete Lyapunov equation.

3 Results

The pattern: a delayed feedback, not an instantaneous antagonism. The CCF oscillates rather than decaying: a dominant contemporaneous peak ($\rho(0) \approx +0.6$) and a negative lobe at lags 4–5 quarters (≈ -0.3 in QoQ), returning to zero by lag 6 (Fig. 1). The event study (Fig. 2) shows profits turning down first and an investment–profit gap that builds up ∼1–1.5 years before the trough—the overaccumulation buildup documented by Tapia Granados (2012), here re-confirmed by an independent method. The effective phase is near co-movement (∼10°), not the 90° of LV/FN; fixed-phase oscillators are accordingly refuted, as their CCF is a rigid cosine, not an isolated pulse plus a delayed negative lobe.

The maturation lag is identifiable and stable. Five methods converge on a maturation lag of ≈ 4 quarters: the empirical kernel, the distributed-lag regression, a VAR($p \geq 4$), a hierarchical pooled estimate ($\mu_{\text{pop}} \approx 4.7$–$4.9$), and the differentiable inverse. The Fourier-feature inverse PINN recovers a known synthetic lag with 8.9% error ($4.0 \to 4.36$ quarters) and estimates 3.90 quarters

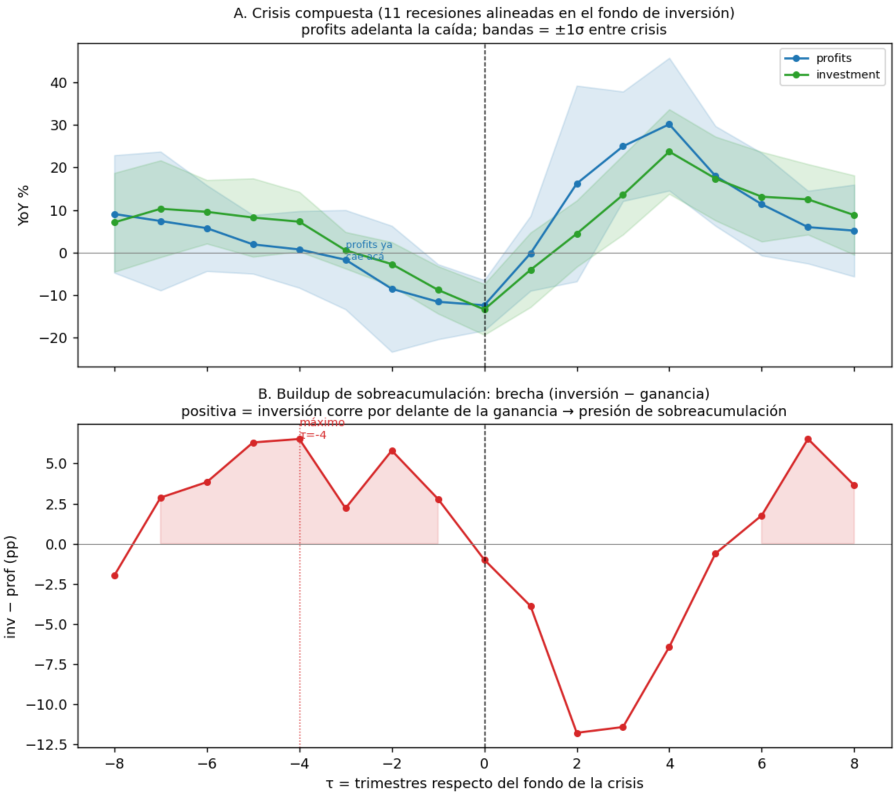

Figure 2: Composite of NBER recessions aligned at the investment trough. Profits lead the downturn and the investment−profit gap builds up ∼1–1.5 years before the trough.

= 0.98 years on real data; the lag is stable across difference transforms (QoQ, $\Delta\log$) and to leave-one-crisis-out (span 0.62 quarter). The classical single-shooting and the latent-state inverse fail (lag collapses or biases by $\sim$55%); the well-posed convolution form with Fourier features succeeds.

What cannot be identified. The feedback explains only $\sim$3% of the variance of profit growth (vs. $\sim$35% for the contemporaneous co-movement). Consequently the *width* of the delay distribution is not identifiable (flat likelihood profile; AIC favours a point delay), and neither is its variation across crises. Moreover, the negative lobe is reproduced by the theoretical CCF of a linear VAR($p \geq 4$) from its coefficients alone (Fig. 3): the distributed-delay kernel is *observationally equivalent*, at the level of the second-order autocovariance, to a linear stochastic system with longer memory. The maturation model is an interpretable reparameterisation of that linear content, not a mechanism separable from it with these data. A bifurcation analysis of the calibrated system does not yield a delay-induced Hopf instability, so we do not claim crises to be an endogenous delayed-feedback instability.

4 Discussion

The substantive picture is consistent and honest. There *is* an endogenous overaccumulation feedback with a maturation lag of about one year, stronger near recessions, and we recover it with a physical, differentiable model where classical inversion fails. But two limits bound the claim. First, the economic content of the lag—its ∼1-year centre and pre-recession buildup—is already in Tapia Granados (2012); our addition is the model and the inverse method, not the phenomenon. Second, with nine usable post-war crises and a $\sim$3% signal-to-variance ratio, the data cannot separate a genuinely *distributed* (stochastic) delay from a point delay, nor the maturation mechanism from a linear long-memory VAR. The microfoundation of a distributed delay is the aggregation of hetero-

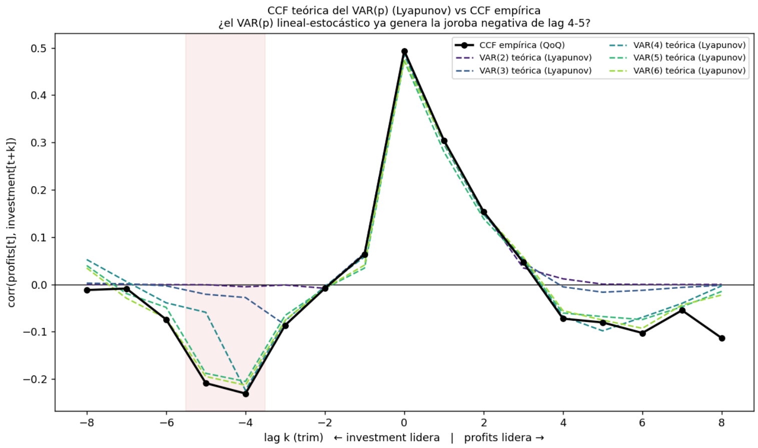

Figure 3: Theoretical CCF of a fitted VAR(p) (Lyapunov) vs. the empirical CCF. For $p \geq 4$ the linear model already reproduces the negative lobe at lags 4–5 from its coefficients, establishing observational equivalence with the distributed-delay kernel.

geneous investment gestation lags (Kalecki, 1935; Kydland and Prescott, 1982), so a distributed kernel is the mechanical consequence of aggregation rather than a conjecture; but confirming its *shape* requires either a multi-country panel (many more episodes, to identify the width) or measured sectoral gestation times (to predict the macro kernel, a test a reduced-form VAR cannot pass). These are the natural next steps.

5 Conclusion

The profit–investment overaccumulation feedback is well described by a distributed maturation delay centred at ∼1 year, recovered by a differentiable inverse that overcomes the spectral bias and ill-posedness of classical calibration. The robust, transferable result is the *location* of the identifiability frontier: the delay's centre is recoverable and stable, while its width and its separation from a linear model are not—a boundary set by signal strength and episode count, not by tuning. Pushing past it requires new data, not a reparameterisation.

A Robustness and falsification battery

Table 1 collects the auxiliary tests. Two points are worth stressing. First, the *location* of the lag (≈ 5 quarters) is stable within the difference family (QoQ, $\Delta \log$) and to leave-one-crisis-out (Fig. 4, left; pooled $\mu \approx 4.7$, span 0.62), although per-crisis confidence intervals are wide because the feedback is weak and identification emerges only by pooling. Second, the negative lobe is *not* a clean crisis-specific signature: a kernel placebo over non-crisis windows finds it roughly as often off-crisis, and a VAR($p \geq 4$) reproduces it from its coefficients (main text, Fig. 3). The year-over-year transform both inflates the magnitude $\sim 2\times$ and shifts the apparent lag (to ~ 2 quarters), and two-sided cycle filters (HP, band-pass) collapse it to a contemporaneous term—so we report QoQ throughout (Fig. 4, right).

B Inverse-PINN variants and a stability check

The maturation lag is recoverable by the differentiable inverse only when the surrogate network can represent the series and the parameters are well-posed. Table 2 compares four variants on synthetic data (true lag 4 quarters) and on the real window. A plain network on the long window suffers spectral bias and the lag collapses; per-crisis pooling fixes the fit ($\bar{R}^2 = 0.86$) but not the lag; only *Fourier-feature* inputs recover the lag cleanly (8.9% synthetic error, 0.98 years real). Finally,

Test	Question	Result
Transform robustness	lag location robust to transform?	$\mu^\star\approx 5$q (QoQ/$\Delta\log$); YoY$\to$2q; filters$\to$1q
Lag stability / LOO	lag stable across crises?	pooled $\mu\approx 4.7$q; LOO span 0.62q; all in CI
Hierarchical pooling	population lag & spread	$\mu_{\mathrm{pop}}\approx 4.9$q $[4.4, 5.5]$; spread contested
Kernel placebo	is the 4–5q lobe crisis-specific?	ubiquitous (appears off-crisis too)
VAR(p) Lyapunov	does a linear VAR reproduce it?	yes for $p\geq 4 \Rightarrow$ obs. equivalence
Pooled-dynamics placebo	is shared crisis dynamics special?	preliminary: ratios $\approx$ crisis $\Rightarrow$ not special

Table 1: Auxiliary robustness and falsification tests. The lag's centre is robust; its crisis-specificity and its separation from a linear model are not.

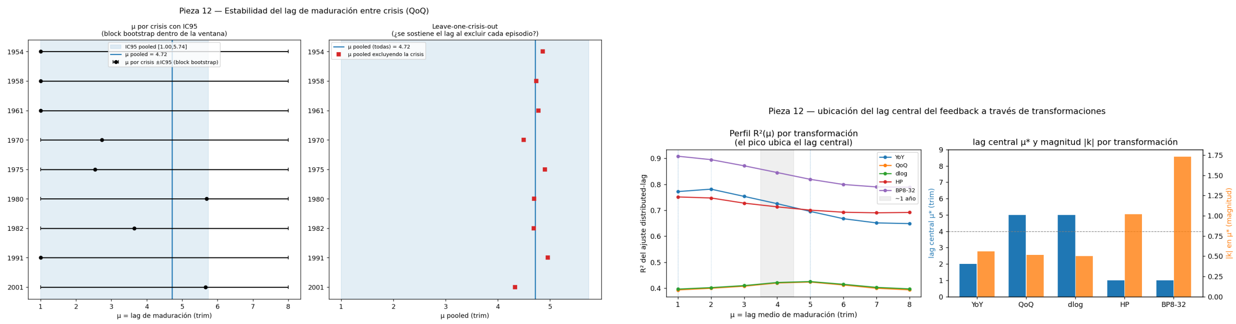

Figure 4: Left: per-crisis maturation lag with leave-one-crisis-out (the pooled lag does not depend on any single episode). Right: estimated lag location under different transforms; stable within the difference family, shifted by YoY and by two-sided cycle filters.

a bifurcation analysis of the calibrated system (Fig. 5) finds *no* delay-induced Hopf instability: the system's instability comes from low damping, not from the lag, and the structural fit even disagrees with the distributed-lag estimate on μ. We therefore do not advance an "endogenous delayed-feedback instability" claim—it would require the stronger identification that a panel or micro gestation data could provide.

Variant	Synthetic (true 4q)	Real lag	Verdict
Single-shooting / convolution	1.86q (54% err)	collapses to 0.23q, non-physical	fails
Per-crisis pooled	1.64q (59% err)	0.84q ($\bar{R}^2=0.86$)	fits, lag collapses
Profile-likelihood in μ	—	inconclusive	unstable
Fourier features	**4.36q** (**8.9%**)	**3.90q= 0.98yr** ($R^2=1$)	**recovers**

Table 2: Inverse-PINN variants. Only Fourier-feature inputs both fit the series and recover the maturation lag; the others fail through spectral bias or the k–μ trade-off.

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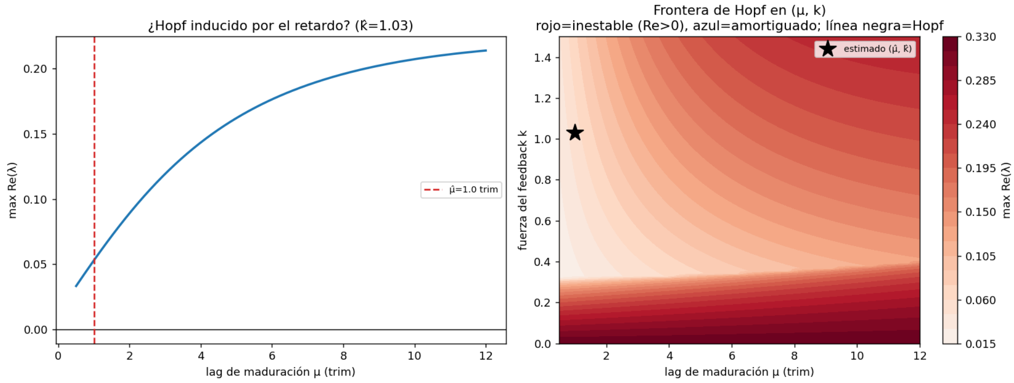

Figure 5: Stability of the calibrated maturation system. Left: maximum $\mathrm{Re}(\lambda)$ vs the lag μ; no zero-crossing, i.e. no delay-induced Hopf. Right: the Hopf boundary in (μ, k) with the estimate marked. The mechanistic "instability" reading is not supported by these data.

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